

EX .5

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
```

```
# Load dataset
```

```
iris = load_iris()
```

```
X = iris.data
```

```
y = iris.target
```

```
# Split data
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
# Create KNN model
```

```
model = KNeighborsClassifier(n_neighbors=3)
```

```
# Train
```

```
model.fit(X_train, y_train)
```

```
# Prediction
```

```
y_pred = model.predict(X_test)
```

```
print("Predicted:", y_pred)
```

```
print("Actual :", y_test)
```

```
print("Accuracy :", accuracy_score(y_test, y_pred))
```

Output

```
*** Predicted: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
Actual   : [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
Accuracy : 1.0
```